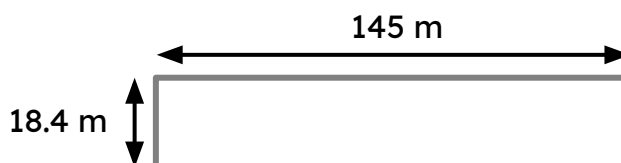




Checking by estimating

Task A: Estimating answers

1) A rectangle has dimensions 145 m by 18.4 m.



Which calculation would give a suitable estimate for the area of the rectangle? Explain why you think it is a suitable estimation.

- a) 150×18 This is suitable because each number has been rounded to 2 s.f.
- b) 150×20 This is suitable because each number has been rounded to the nearest 10
- c) 100×20 This is suitable because each number has been rounded to 1 s.f.
- d) 18×145 This is suitable because each number has been rounded to the nearest integer.

2) Here are 3 damaged receipts. Each pupil pays with a £20.

Aisha gets £12.23 change. Jacob gets £6.89 change. Sam gets £9.11 change. Only using estimation, which student went to which shop?

Shop A	
Items	Price
Calc	£5.12
Pencil	£0.78
Ruler	£1.89
Paper	£3.10
Total	

Estimating
 $£5 + £1 + £2 + £3 = £11$
 Approximate change
 expected £9.
 Sam went to Shop A

Shop B	
Items	Price
Pencil	£1.02
Stamps	£7.83
Label	£2.99
Crisps	£1.27
Total	

Estimating
 $£1 + £8 + £3 + £1 = £13$
 Approximate change
 expected £7.
 Jacob went to Shop B

Shop C	
Items	Price
USB	£5.03
Paper	£0.28
Pens	£0.91
Pencil	£0.56
Crisps	£0.99
Total	

Estimating
 $£5 + £0.30 + £1 + £0.50 +$
 $£1 = £7.80$
 Approximate change
 expected £12.20.
 Aisha went to Shop C

3) Another receipt was damaged. Jun remembers paying with a £50 note and receiving about £3 back in change. Estimate how much the board game was. You must show your working out.

Items	Price
Stamps	£5.31
Calc	£8.12
Pencil	£0.94
Ruler	£1.89
Paper	£4.50
USB	£8.59
B. Game	£
Total	

Estimating

$$£5 + £8 + £1 + £2 + £5 + £9 = £30$$

Given the change from £50 was £3

$$£50 - £33 = £17$$

The board game was approximately £17

Task B: Checking answers

1) Using estimation only, put a tick with the calculation which gives the approximate correct answer. Where the answer is incorrect, identify the error.

Buying 34 books each costing £21.32 $\approx$ £700

$$35 \times 20 = 700$$



A rectangle with lengths 3.21 cm by 9.56 cm has an area $\approx$ 30 cm²

$$\text{Area} = 3 \times 10 = 30$$



A square with length 8.23 mm has a perimeter $\approx$ 64 mm

$$\text{Area} = 8 \times 8 = 64$$

$$\text{Perimeter} = 8 + 8 + 8 + 8 = 32$$

There are approximately forty five 50p pieces in £91.

$$45 \times 0.5 = 45 \div 2 = 22.5$$

$$\text{Or } 90 \div 0.5 = 90 \times 2 = 180$$

2)

$$7.03 \times \frac{5}{11}$$

Is approximately:

6

14

3

4

Not possible to estimate

Make sure to show your reasoning or working out.

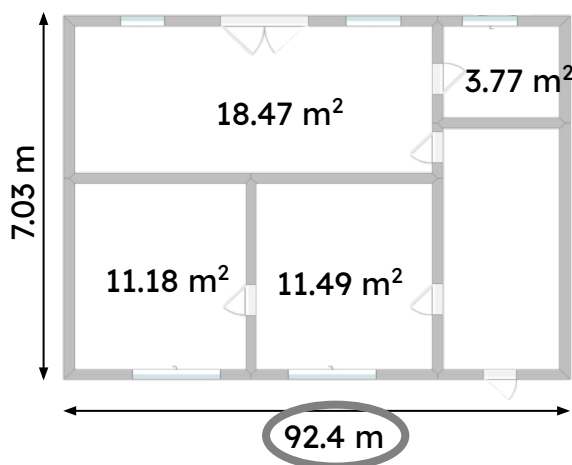
There are a few ways to determine this approximation.

Example, $\frac{5}{11}$ is slightly below a half so a little below half of 7 is 3.

Example is rounding $7 \times \frac{5}{11} = \frac{35}{11} \approx \frac{33}{11}$ giving an approximate answer of 3



3) Here is a floor plan of a one bedroom apartment. The Oak teacher spots an error with a measurement given.



a) Can you find the error? Estimate what the correct measurement should be.

9 m. Accept anything between 8 m and 12 m inclusive.

b) Can you estimate the area of the missing room?

Estimating the length to be 9 m, the total area is expected to be $7 \times 9 = 63$.

Therefore, rounding and subtracting the other areas
 $63 - 20 - 4 - 11 - 11 = 17 \text{ m}^2$.

Accept anything between 15 m² and 20 m² inclusive.

